Chapter

I

Number Systems

Section 1. Introduction and Motivation

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Section 3. Properties of the Real Numbers

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1 Introduction and Motivation

The realm of real analysis, born in the early 19th century, ignited a transformative spark that revolutionised the landscape of modern mathematics. This monumental shift finds a parallel in the trajectory of a typical undergraduate mathematical journey, where real analysis, and perhaps linear algebra, stands as the initial encounter demanding proof and rigour above mere calculations. Such a juncture often brews bewilderment and even aversion among mathematicians, much akin to the sentiments experienced by some of the greatest mathematical minds when embracing this novel perspective. This kind of discomfort is often attributed to the axiomatic nature that real analysis favours – a departure from the algorithmic and intuitive mathematical terrain previously trodden. Progress is definitely slower using this axiomatic approach, and arguably intuition is lost, yet, due to its rigour, we can be assured that progress is definitely made and no mathematical contradictions may appear. It is due to this that it is this form of mathematics that has prevailed and morphed into the modern mathematics that is done throughout the world today.

In is worth commenting though that the subject of real analysis doesn't exist purely for the sake of rigour, despite how it may come across and how it is portrayed in many textbooks. In fact, if it did, it wouldn't have caught on as it has. In reality, rigour was a necessity in order to solve complicated problems as well as resolve apparent contradictions relating to the nature of infinity and in turn, calculus. As we embark on this journey, our starting point is to chart a course through the challenges that once rattled the very foundations of mathematics – particularly the realm of calculus. To fully comprehend why we delve into the bedrock of mathematics itself, even peering into the essence of numbers, a robust motivation is indispensable.

Problem 1 - The nature of infinite summations

The term *infinite summations* is an oxymoron. The word *infinite* means without limit and never ending whilst the word *summation* means the act of coming to the highest point, rooting from the word *summus* or summit. How can it make sense then to come to an end of a process that never ends? This already hints that we must take great care when defining the term *infinite series*. For now, by *infinite series* we will mean a summation that has no end.

Historically, people had been working with infinite series in a rather ad hoc manner and in the 17th and 18th century they treated them as finite summations with a very large number of terms. People were well aware at the time though of the distinction between series with a large number of summands and infinite series. They knew that serious problems could arise if this distinction wasn't made. Still, the appeal towards treating infinite series as if they really were summations was enchanting since it led to useful practical insights such as the integral of complicated

functions. For example, treating integration as the inverse of differentiation, there would be great difficulty in integrating the function e^{-x^2}. Instead, since

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \dots,$$

where we think of each sum here in the usual way and is simply added indefinitely, we can integrate this function by integrating term by term. Providing we stay within the region in which this sum *converges* to e^{-x^2}, no problems arise. So if it is the belief that infinite series really are just a continuation of usual addition, then surely the usual properties of addition should remain within an infinite series, especially if that series converges to a specific value or function. Let's explore this hypothesis through an explicit example. It can rather unequivocally argued that

$$1 + 2 = 2 + 1.$$

This is known as the *commutativity* of addition. Now, consider the infinite series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots.$$

Does this series get closer and closer to one particular value as more terms are added? Summing the first 100 terms of this expression gives us a rough value of 0.688. Summing the first 500 terms gives us a rough value of 0.692. So it does indeed look like this sum is *converging* to a specific value. Surely if we simply rearrange this infinite series using the commutativity of addition it should give the same value. Let's test this. Rearrange the infinite series as

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots.$$

If infinite series truly are infinite summations, and summations are commutative, this should be equal to the previous series. Summing the first 100 terms of this series gives an approximate value of 1.032. Perhaps we just need to sum more terms. Summing the first 500 terms gives an approximate value of 1.038. Now the infinite series appears to be converging to a different value. What is going on? How can it be that the same infinite series is converging to two different numbers? It appears then that we cannot treat all infinite series as true summations. The natural question arises, which infinite series can we treat in this way and why? In order to answer this we need a rigorous definition of infinite series which we will arrive at in this course.

Problem 2 - Fourier Series

Whilst the first problem with infinite series was beginning to rock the foundations of mathematics, it was the work of a young mathematician by the name of Joseph Fourier that shook the very edifice of calculus. Fourier was studying the propagation of heat in solid bodies. That is, he wanted to know how heat flows through solid bodies. To simplify the problem he considered an simpler body in the shape of a very long and very thin rectangular plate. He also considered the idealised situation in which the body lost no heat to its surroundings and the two long sides

of the sheet were held at constant temperature, which he set to 0. The sheet can be thought of as lying in the x, y plane with region bounded below by the x-axis. It is taking that the left hand side of the sheet is at $x = -1$ and the right and side is at $x = 1$ and the temperature of the body is given by $T(x, y)$. As outlined above, it is chosen so that the temperature satisfies

$$T(-1, y) = T(1, y), \text{ for } y > 0.$$

With this setup, Fourier wanted to know what the temperature distribution was throughout the lamina given an initial temperature distribution $f(x) = T(x, 0)$. The question remained though as to what $T(-1, 0)$ and $T(1, 0)$ should be. By considering situations in which the temperature drops off to zero as x approaches 1 along the bottom edge, he managed to find a solution for $T(x, y)$. Specifically he found that if the original temperature distribution $f(x)$ could be written as

$$\begin{aligned} f(x) = a_1 \cos\left(\frac{\pi x}{2}\right) + a_2 \cos\left(\frac{3\pi x}{2}\right) + a_3 \cos\left(\frac{5\pi x}{2}\right) + \dots \\ + a_n \cos\left(\frac{(2n-1)\pi x}{2}\right), \end{aligned} \tag{1.1}$$

where $a_i \in \mathbb{R}$ for each $i \in \mathbb{R}$, then

$$\begin{aligned} T(x, y) = a_1 e^{-\pi y/2} \cos\left(\frac{\pi x}{2}\right) + a_2 e^{-3\pi y/2} \cos\left(\frac{3\pi x}{2}\right) + \dots \\ + a_n e^{-(2n-1)\pi y/2} \cos\left(\frac{(2n-1)\pi x}{2}\right). \end{aligned}$$

That is, the temperature of the sheet drops off exponentially as you move away from the x-axis. The follow question now emerges. What if the original temperature distribution cannot be written in the form of (1.1)? For example, consider the physical situation in which the $f(x) = 1$ for $-1 < x < 1$. This is simply the situation in which the temperature distribution is constant, something that is entirely plausible physically. But, the constant temperature distribution cannot be written in the form of (1.1) since this function is discontinuous at $x = 1$ and (1.1) implies that $f(x)$ must be continuous at $x = 1$. This didn't sit well with Fourier. After all, it is possible to apply a constant temperature to one end of a metal bar. He noticed that if larger and larger values are taken for n in (1.1), and some suitable choices for $a_i \in \mathbb{R}$ are made, this function more closely approximates $f(x) = 1$. He boldy claimed that by taking infinitely many terms, and his choices for $a_i \in \mathbb{R}$, we get a function that exactly equals $f(x) = 1$. That is

$$f(x) = \frac{4}{\pi}\left(\cos\left(\frac{\pi x}{2}\right) - \frac{1}{3}\cos\left(\frac{3\pi x}{2}\right) + \frac{1}{5}\cos\left(\frac{5\pi x}{2}\right) + \dots\right) = 1$$

for $-1 < x < 1$ and thus

$$T(x, y) = \frac{4}{\pi}\left(e^{-\pi y/2}\cos\left(\frac{\pi x}{2}\right) - \frac{1}{3}e^{-3\pi y/2}\cos\left(\frac{3\pi x}{2}\right) + \dots\right) \quad .$$

If this were true, it would shake the very foundation of what it even means for something to be a function, at least as the word *function* was understood at the

time. At the time, functions were understood as concrete formulas such as $\sqrt{x}$, x^2 or $\sin(x)$. They were inherently differentiable and continuous. It was thought that if the value of a function was known within a region, then the value of a function could be determined everywhere. But if Fourier's claim was correct, this would not be the case, since $f(x) \neq 1$ for all $x \in \mathbb{R}$ and even worse, it isn't even continuous. People argued that Fourier's claim must be wrong and that this infinite series cannot possibly converge to 1 for $-1 < x < 1$. Regardless of the mathematics, the problem is the temperature distribution could be measured physically and it agreed with Fourier's solution. Fourier also went on to produce a range of solutions that could be verified physically under the assumption that any initial temperature distribution could be written as an infinite series of sines and cosines. It was later proven by Fourier and others that this infinite series does indeed converge to 1, but to do this, it forced the mathematical community to critically re-evaluate what it meant by a function, an infinite series and a derivative. As each component of calculus came under scrutiny, it was found to rest on uncertain foundations that needed to examined and reconstructed.

This is be no means an exhaustive list of problems that led to the development of what is now known as real analysis, or just analysis. What these problems outline though is that when dealing with the infinite, strange and paradoxical mathematical statements can emerge. Further, these problems could not be simply argued away. The current calculational techniques that had worked so effectively since the development of calculus were insufficient for rectifying such problems. It could be argued that these are just mathematical curiosities, with no application in the 'real' world and therefore such rigour is not needed for solving so called 'real world problems'. This argument doesn't hold much water though, given the intimate relationship between Fourier series and solutions to the equations that describe the dynamics of our physical reality, namely partial differential equations. We are then faced with the harsh realisation that something needs to be corrected at the very foundation in the understanding of infinity, infinitesimals and the real numbers themselves. This leads naturally to the opening section.

2 What is a number?

This question is perhaps bordering more on the philosophical however, we will not dive down that rabbit hole and instead start from what is perhaps the most natural way of describing a number, that is as a representation of a unit of measure. Like most things in mathematics, we should aim to start with the most basic structures and build from there. It seems logical then to start with the most basic units of measure, namely discrete ones. That is, the ones that count objects. Quite aptly, these are the very same you start with in primary school. In university, we call this collection of numbers, the *natural numbers*.

Definition 2.1: Natural Numbers

The set of ***natural numbers***, denoted by $\mathbb{N}$, are given by

$$\mathbb{N} := \{1, 2, 3, \dots\}.$$

With the natural numbers being measures to count objects, it is sensible to ask are there numbers that count the absence of objects, otherwise known as debt. As you might expect, these are precisely the *negative integers* along with zero. Grouping these together with the natural numbers defines the *integers*.

Definition 2.2: Integers

The set of ***integers***, denoted by $\mathbb{Z}$, are given by

$$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Continuing along with this story, the next measure would be ratios of integers. That is the numbers that quantify fractional wholes. These are more commonly referred to as the *rational numbers* and are defined as follows.

Definition 2.3: Rational Numbers

The set of ***rational numbers***, denoted by $\mathbb{Q}$, are given by

$$\mathbb{Q} := \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}.$$

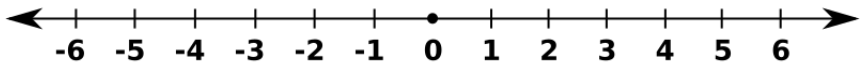
So far we have only presented numbers as representations to specific measurements with no discussion as to the possible relationships between individual numbers, and in turn measurements. For example, if the naturals represent the counting of objects, say apples, it is quite reasonable to say that one apple plus another is the same as two. In other words, a mathematical operation can be defined on $\mathbb{N}$, commonly known as *addition*. Equally, by similar arguments, the distinct operation of *multiplication* can defined on the natural numbers. Both of these operations can then be extended to the rational numbers so that multiplication and addition of fractions is defined in the usual manner. In mathematical terminology, we have defined two *binary operations* on $\mathbb{Q}$, namely addition $(p, q) \mapsto p + q$ and multiplication $(p, q) \mapsto p \cdot q$. This isn't the end of the structures that can be defined on our number systems though. Again, returning to the idea of the natural numbers representing the counting of objects, we also have a notion of one number being larger than another. Mathematically, this is known as an *ordering* on a set and is denoted by the symbol $<$. Like addition and multiplication, this ordering can be extended to be defined on $\mathbb{Q}$ in the usual way. With these operations and relations in mind, we now view $\mathbb{Q}$ as an *algebraic structure* meaning that it is the combination of all the fractional numbers as well as the usual ordering and operations of addition and multiplication. Here, the term algebraic structure simply means that both the set and the relations/operations defined on the set are considered collectively.

2.1 Who ordered these?

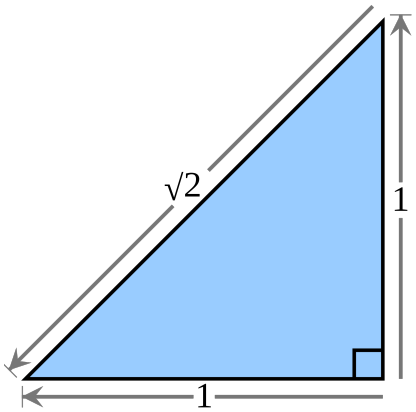
Unlike $\mathbb{N}$ and $\mathbb{Z}$, the rational numbers have the special property that between any distinct $x, y \in \mathbb{Q}$ there exists $z \in \mathbb{Q}$ such that $x < z < y$. This is clear by defining

$$z = \frac{y + x}{2}.$$

Given this property, it seems quite reasonable to represent rational numbers by a straight continuous line extending infinitely from left to right, as seen in the following picture.



If this were the case, then it would suggest that the rational numbers are sufficient for representing the length of all line segments. In other words, we could conjecture that given any two line segments, there always exists a third line segment that divides evenly into the first two. With this in mind, consider the following picture:



Now if this conjecture were true, this would mean that the length of the hypotenuse of the triangle would be a rational multiple of either of the other two sides. That is to say, that the square root of two is a rational number. This now leads us to arguably one of the most profound proofs in mathematics.

Theorem 2.4: Irrationality of $\sqrt{2}$

The $\sqrt{2}$ is not rational.

Proof. Suppose towards a contradiction that $0 < \sqrt{2}$ is rational. That is, $\sqrt{2} \in \mathbb{Q}$. Hence, there exists some $a, b \in \mathbb{N}$ such that $\sqrt{2} = \frac{a}{b}$. Assume without the loss of generality that $\frac{a}{b}$ is irreducible. That is, there are no common divisors of a and b other than 1. It follows that $a^2 = 2b^2$ and hence a^2 is even. Now, if a^2 is even, then a is also even. Given this, it must be that $4 \mid 2b^2$ and hence b^2 is even. But, if $2 \mid b^2$ this implies $2 \mid b$. As $2 \mid a$ and $2 \mid b$, it follows that $\frac{a}{b}$ is reducible which is a contradiction. □

Whilst this proof is rather elegant, we must now face the hard truth that the rational numbers are not sufficient to represent all lengths. Rather than abandon

arithmetic in favour a geometry, as the Greeks did, we plan to extend, or complete, the rational numbers so as to be able to assign numerical values to any length.

To this end, we start by identifying all the properties of the rational numbers, as whatever this new number system will be, should it even exist, must minimally satisfies all the same properties of the rational numbers, plus some more. This leads to the following definition.

Definition 2.5: Ordered Field

An ***ordered field*** $\mathbb{F}$ is a non-empty set with two binary operations, addition $(+)$ and multiplication $(\cdot)$, as well as two relations $(<)$ and $(=)$, such that

(i) (Commutative Law) If $a, b \in \mathbb{F}$, then $a + b = b + a$ and $a \cdot b = b \cdot a$.

(ii) (Distributive Law) If $(a, b, c) \in \mathbb{F}$, then $a \cdot (b + c) = a \cdot b + a \cdot c$.

(iii) (Associative Law) If $a, b, c \in \mathbb{F}$, then $a + (b + c) = (a + b) + c$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

(iv) (Identity Law) There exists elements $0, 1 \in \mathbb{F}$ such that $a + 0 = a$ and $a \cdot 1 = a$ for all $a \in \mathbb{F}$.

(v) (Inverse Law) For every $a \in \mathbb{F}$, there exists $-a \in \mathbb{F}$ such that $a+(-a) = 0$. If $a \neq 0$, there also exits an element $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$.

(vi) (Order Axioms) Take $a, b \in \mathbb{F}$.

- (a) Exactly one of the relations $a < b$, $b < a$ or $a = b$ holds.
- (b) If $a < b$, then for every $c \in \mathbb{F}$ it follows that $a + c < b + c$.
- (c) If $0 < a$ and $0 < b$, then $0 < a \cdot b$.
- (d) If $b < a$ and $c < b$, with $c \in \mathbb{F}$, then $c < a$.

A ***field*** is simply an *ordered field* but with the order axioms removed.

This definition may seem rather abstract and detached from our so far fairly grounded discussion relating measurement to numbers. But, it should be clear where these axioms are coming from in that they emerged as a result of the algebraic properties of the usual structures defined on $\mathbb{Q}$. You may wonder though why to bother to do this at all. Why not simply work with $\mathbb{Q}$ as opposed to this general concept of an ordered field. It is here where we start to see a glimpse into this paradigm shift that changed the way in which modern mathematics is done. Namely, the process of *abstractifying* concrete mathematical examples. What is meant here by abstractifying or abstraction? In this context, abstraction is the process by which we remove irrelevant details to expose an underlying structure. To motivate this, consider the following. So far, we have discussed numbers as representations of measurement, say length. We realise though that there is an arbitrary nature to the way that length is measured. It could be choosen so that the fundamental unit of measurement is meters, centimetres, feet, yards, etc. If it was chosen so that a meter stick represented the number 1, another could equally choose for the yard stick to represent the number 1. In each case, valid measurements are

produced by utilising the addition, multiplication and ordering of numbers, even though the representations of the number 1 are given by different physical lengths. So it really is that measurements are representations of some deeper mathematical structure as opposed to the other way around. What this emphasises is that it isn't the specific elements of a set that you care about. Instead, a mathematician only cares about the underlying structures on the set and the properties they satisfy. In this case, ordering, addition and multiplication. It is then the goal of the mathematician to strip away all of the unnecessary detail to reveal the underlying deeper structure. This is the greatest weapon to any trained mathematician.

2.2 Filling in the gaps

We have now seen that ordered fields are insufficient for describing continuous quantities since ordered fields like $\mathbb{Q}$ contain infinitely many holes. What is the property then needed so as to fill in all of these gaps? To answer this, it is fruitful to return back to the rational numbers and think about where exactly these gaps are in relation to the rational numbers. We now know that $\sqrt{2}$ is not a rational number. What we can say for sure though is that it is bounded above and below by two rational numbers, say 2 and 1. Since between any two distinct rational numbers there exists a rational number, the interval that $\sqrt{2}$ is bounded by can be shrunk indefinitely and therefore produce better rational approximations to $\sqrt{2}$. This hints to us that the gaps in the rational numbers are between all the rational numbers. Before moving further, it will be useful to introduce some terminology.

Definition 2.6: Bounded Set

Let S be a non-empty subset of an ordered field $\mathbb{F}$. Then S is said to be ***bounded above*** if there exists a $\alpha \in \mathbb{F}$ such that $x \leq \alpha$ for all $x \in S$. The number α is called an ***upper bound*** for S.

Similarly, the set S is said to be ***bounded below*** if there exists a $\beta \in \mathbb{F}$ such that $\beta \leq x$ for all $x \in S$. The number β is called a ***lower bound*** for S.

The set S is said to be ***bounded*** if it is bounded both above and below.

With this definition in mind, consider the subset

$$A = \{x \in \mathbb{Q} : x > 0, x^2 < 2\} \subset \mathbb{Q}.$$

It is clear that this set is bounded both above and below. Note that there are infinitely many values that bound this set from above and below. In the definition of bounded it doesn't say anything about the bounding element being unique. For example, in the case of A, it is bounded above by $4, 5/2, 2$ and many more. Suppose now you collect all the possible upper bounds for the set A into a new set, say A^u. Does this set have a smallest element? After some thought, we realise that the answer is no as this element when squared would have to equal 2, which wouldn't

be rational. So what would happen if we only considered ordered fields for which whenever we take subsets of them, that are bounded above, that the collection of upper bounds has a least element? This is precisely the condition we need so as to fill in all the gaps in the rational numbers and leads us to a formal definition of *completeness*. We will see in detail in a following section how this property allows for the existence of square roots. Before this, one more piece of jargon is needed to arrive at the definition of completeness.

Definition 2.7: Supremum and Infimum

Let A be a non-empty subset of an ordered field $\mathbb{F}$ that is bounded above. Then the ***supremum*** of A, should it exist, is the least upper bound of A and is denoted by $\sup(A)$. That is, if $a \in \mathbb{F}$ is an upper bound for A, then $\sup(A) \leq a$.

Similarly, if B is a non-empty subset of an ordered field $\mathbb{F}$, that is bounded below, the ***infimum*** of B, should it exist, is the greatest lower bound of B and is denoted by $\inf(B)$. That is, if $b \in \mathbb{F}$ is a lower bound for B, then $b \leq \inf(B)$.

Definition 2.8: Complete Field

An ordered field $\mathbb{F}$ is said to be ***complete*** if for every non-empty subset $A \subset \mathbb{F}$ bounded above, there exists a supremum in $\mathbb{F}$

Equally, completeness could be defined as every non-empty subset $A \subset \mathbb{F}$ bounded below has a infimum in $\mathbb{F}$. Proving the equivalence of these two statements will be an exercise. With these definitions in place, we are finally in a position to introduce the real numbers.

Theorem 2.9: Real Numbers

There exists a unique complete ordered field $\mathbb{R}$. This field is called the ***real numbers.***

This theorem will be stated without proof. This is not just for the sake of brevity, but also the nature of this proof is not particularly enlightening to the rest of the course. It is also here that the advantages of the axiomatic approach to mathematics begin to emerge. From this theorem, we are able to deduce that every complete ordered field is the same, or more accurately it is isomorphic, to the real numbers. This is quite a profound statement as what it is saying is that the real numbers hold some special role unlike other number systems. It is in fact the only complete ordered field that can be produced. Mathematically, the real numbers are said to satisfy a *universal property*. This is not a concept that will be explored in detail within this course, but it is worth mentioning that many mathematical structures satisfy so called universal properties and gives hints to even deeper structures underlying all of mathematics.
Before moving on it will be fruitful to wrap up this section by unravelling this

theorem so as to explicitly identify the properties that the real numbers satisfy.

Corollary 2.10: Real Numbers

The real numbers $\mathbb{R}$ are a non-empty set with two binary operations, addition $(+)$ and multiplication $(\cdot)$ such that

(i) (Commutative Law) If $a, b \in \mathbb{R}$, then $a + b = b + a$ and $a \cdot b = b \cdot a$.

(ii) (Distributive Law) If $(a, b, c) \in \mathbb{R}$, then $a \cdot (b + c) = a \cdot b + a \cdot c$.

(iii) (Associative Law) If $a, b, c \in \mathbb{R}$, then $a + (b + c) = (a + b) + c$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

(iv) (Identity Law) There exists elements $0, 1 \in \mathbb{R}$ such that $a + 0 = a$ and $a \cdot 1 = a$ for all $a \in \mathbb{R}$.

(v) (Inverse Law) For every $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a+(-a) = 0$. If $a \neq 0$, there also exits an element $a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$.

(vi) (Order Axioms) Take $a, b \in \mathbb{R}$.

- (a) Exactly one of the relations $a < b$, $b < a$ or $a = b$ holds.
- (b) If $a < b$, then for every $c \in \mathbb{R}$ it follows that $a + c < b + c$.
- (c) If $0 < a$ and $0 < b$, then $0 < a \cdot b$.
- (d) If $b < a$ and $c < b$, with $c \in \mathbb{R}$, then $c < a$.

(vii) (Completeness Axiom) Given a non-empty set $A \subset \mathbb{R}$ that is bounded above, then $\sup(A) \in \mathbb{R}$.

For brevity, and my own sanity, the multiplication of a and b will be written as ab as opposed to $a \cdot b$ throughout the text.

Notice that unlike the definition of an ordered field, $\mathbb{R}$ satisfies the completeness axiom which is the property that makes it distinct from $\mathbb{Q}$. In the next section we will see what are the direct consequences of the completeness axiom. Finally, it is important to emphasise that every result proven in this course will result purely from these axioms here. This is fundamentally how modern mathematics is done. That is, the enormity of theorems and lemmas proven in any mathematical discipline come from the fewest number of assumptions, or axioms, that can be made.

3 Properties of the Real Numbers

With the definition of the real numbers finally in hand, we are now able to prove some properties about them. We start by observing that given that $\mathbb{R}$ is an ordered field, we are able to define a notion of distance between two real numbers. This will be crucial when discussing notions of convergence, or 'closeness', in later chapters.

Definition 3.1: Modulus

The ***absolute value function*** $|\cdot| : \mathbb{R} \to \mathbb{R}$ is defined as

$$|x| = \begin{cases} x : x \geq 0 \\ -x : x < 0 \end{cases}.$$

Example 3.2: Modulus

(i) $|7| = 7$.

(ii) $|-5| = -(-5) = 5$.

(iii) $|0| = 0$.

The absolute value function satisfies a lot of properties, each of which can be crucial for certain proofs. All of these properties will be outlined here, but not all will be proven. We will only prove the *triangle inequality* here and the rest are left as exercises.

Theorem 3.3: Properties of Absolute Value Function

(i) $|a| \geq 0$, with equality if and only if $a = 0$.

(ii) $|a| = |-a|$.

(iii) $-|a| \leq a \leq |a|$.

(iv) $|ab| = |a||b|$.

(v) $1/|a| = |1/a|$, if $a \neq 0$.

(vi) $|a|/|b| = |a/b|$, if $b \neq 0$.

(vii) $|a| \leq b$ if and only if $-b \leq a \leq b$.

(viii) (Triangle Inequality) $|a+b| \leq |a| + |b|$.

(ix) (Reverse Triangle Inequality) $||a| - |b|| \leq |a-b|$.

Proof. (viii) For all $a, b \in \mathbb{R}$ it follows that

$$a + b \leq |a| + |b|.$$

Similarly,

$$-(a+b) = (-a) + (-b) \leq |-a| + |-b| = |a| + |b|$$

by (ii). By the definition of the absolute value function it follows that

$$|a+b| \leq |a| + |b|$$

as required. □

It is worth emphasising that the triangle inequality is perhaps the most exploited inequality in real analysis proofs and thus it is crucial that you remember it. In addition, the key feature that makes real analysis proofs different to say linear algebra is precisely that we are working with ordered fields. Since the very nature of inequalities means that we do not necessarily have equality, there are often infinitely many inequalities that can be written down in an analysis proof. It is frequently required that a suitable inequality is dreamt up so as to simplify the calculations greatly. It is this less algorithmic method of producing proofs and necessity on imagination that often causes confusion/difficulty with analysis proofs.

Note that whilst we may have defined the reals, we actually don't know much about them other then some algebraic and order properties along with the completeness axiom. Indeed the rationals are contained within them but, we don't know precisely what new numbers were added to $\mathbb{Q}$ by the completeness axiom. We ideally would like to picture the real numbers lying on a continuous numberline stretching indefinitely from left to right, but in order to do this me must prove some similar properties of the reals. Let's start with the property that if the reals were represented by a line, this line should extend indefinitely. This is shown by the so called *Archimedean property* of the real numbers.

Theorem 3.4: Archimedean Property

For every $x, y \in \mathbb{R}$ with $0 < x < y$, there exists an $n \in \mathbb{N}$ such that $y < nx$.

Proof. By setting $z = \frac{y}{x}$, the above statement is equivalent to for every $z \in \mathbb{R}$ there exist an $n \in \mathbb{N}$ such $z < n$. Assume towards a contradiction that there exists an $u \in \mathbb{R}$ such that $n \leq u$ for every $n \in \mathbb{N}$. This means that $\mathbb{N}$ is bounded above and hence by the completeness axiom has a least upper bound. Define $\alpha = \sup(\mathbb{N})$. Then $\alpha - 1 < \alpha$ and is not an upper bound for $\mathbb{N}$. Hence, there must exist $m \in \mathbb{N}$ such that

$$\begin{aligned} &\alpha - 1 < m \leq \alpha \\ \Rightarrow\ &\alpha < m + 1 \in \mathbb{N}. \end{aligned}$$

This is a contradiction since α was assumed to be the least upper bound of $\mathbb{N}$. □

This theorem may seem entirely obvious and you may wonder why to prove it at all. The subtlety lies in the completion axiom. All we know so far is that $\mathbb{R}$ is an extension of $\mathbb{Q}$ that satisfies the same algebraic and order properties of the rationals but is also complete. How do we know that we didn't accidentally add some new exotic numbers that have the property that they bound the natural numbers? As crazy as this may sound, there are actually ordered field extensions of $\mathbb{Q}$ that include such objects. Some examples are the *surreal numbers* and the *hyperreal numbers*. These will not be studied here but are mentioned for the interested reader. The Archimedean Property demonstrates that the completeness axiom did not introduce such numbers. Similarly, we can deduce from the Archimedean property that no *infinitesimals* exist in $\mathbb{R}$. This is captured by the following corollary.

Corollary 3.5: Archimedean Property

For every $z \in \mathbb{R}$ with $z > 0$, there exists a $n \in \mathbb{N}$ such that $\frac{1}{n} < z$.

Proof. Follows from Theorem 3.4 by setting $z = x/y$. □

With the Archimedean property in hand we are one step closer to being able to visualise the real numbers all sitting on a continuous numberline. The key property to prove now about the reals is that there are no gaps, unlike the rational numbers. How might we do this? One way is to imagine a sequence of closed intervals on $\mathbb{R}$ can narrower and narrower. If the real truly have no gaps, then these intervals must always contain a real number. This is precisely what is known as the *nested interval property.*

Theorem 3.6: Nested Interval Property

Let $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ be a closed interval with $I_{n+1} \subset I_n$ for each $n \in \mathbb{N}$. Then there exist an $x \in I_n$ for every $n \in \mathbb{N}$.

Proof. Define $A = \{a_n : n \in \mathbb{N}\}$. Since the intervals are nested, that is $I_{n+1} \subset I_n$, it follows that every b_n is an upper bound for A. By the completeness axiom, A has a least upper bound, call this α. Hence, $a_n \leq \alpha \leq b_n$ for every $n \in \mathbb{N}$. So $\alpha \in I_n$ for every $n \in \mathbb{N}$. □

With the nested interval property and the Achimedean property we can now truly represent $\mathbb{R}$ as a continous infinite numberline. We are still yet to see though how the rational numbers sit inside this numberline. Are they possibly clumped in one region or spread out in an even fashion? To see how they fit in exactly, we prove the following theorem.

Theorem 3.7: Density of $\mathbb{Q}$ in $\mathbb{R}$

For every $x, y \in \mathbb{R}$ with $x < y$ there exists a $z \in \mathbb{Q}$ such that $x < z < y$.

Proof. Suppose that $0 < x < y$. Then by the Archimedean property there exists an $n \in \mathbb{N}$ such that

$$\frac{1}{n} < y - x.$$

By the definition of $\mathbb{N} \subset \mathbb{R}$, there exists a smallest element $m \in S$ with

$$S = \{k \in \mathbb{N} : k > nx\}.$$

Hence, $nx < m$. Now, since $m - 1 \notin S$ this means $m - 1 \leq nx \Rightarrow m \leq nx + 1 < ny$. Putting this all together gives

$$x < \frac{m}{n} < y.$$

For the case when $x < 0$ and $y > 0$, simply pick $z = 0 \in \mathbb{Q}$. Finally, when $x, y < 0$ it follows that $-x, -y > 0$. By the above, there exists a $z \in \mathbb{Q}$ such

that $-y < z < -x$ and hence $x < -z < y$. □

Theorem 3.7 can be written succinctly as $\mathbb{Q}$ is *dense* in $\mathbb{R}$. Similarly, it is not hard to prove as a result of Theorem 3.7 that the *irrational numbers*, that is numbers in $\mathbb{R} \setminus \mathbb{Q}$ are also dense in $\mathbb{R}$.

Corollary 3.8: Density of irrationals in $\mathbb{R}$

For every $x, y \in \mathbb{R}$ with $x < y$ there exists a $z \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < z < y$.

Proof. Exercise □

4 Calculating Suprema and Infima

With the properties of the real numbers under our belt, we can begin to calculate some explicit suprema and infima of bounded subsets of the reals. Recall that the key property of the real numbers is that every subset bounded above has a least upper bound, or supremum. Thus we suspect numbers such as $\sqrt{2}$, that were not rational, are now real numbers since they will be suprema, or infima, of some bounded subsets of $\mathbb{R}$. In other words, all the numbers that have been added to the rationals are simply suprema, or infima, of bounded sets. However, what happens if a bounded subset has two distinct suprema? These new numbers could not simply be defined as suprema of bounded subsets since there would be more than one number associated to each supremum. Fortunately, there is a theorem that comes to the rescue.

Theorem 4.1: Uniqueness of Suprema

Let $A \subset \mathbb{R}$ be non-empty and bounded above. Then $\alpha = \sup(A)$ is unique.

Proof. Assume towards a contradiction that there exist two distinct least upper bounds for A, say $\alpha, \alpha' \in \mathbb{R}$. Then since α is a least upper bound for A it follows that $\alpha \leq \alpha'$. Similarly, since α' is a least upper bound for A it follows that $\alpha' \leq \alpha$. Hence, $\alpha = \alpha'$. □

Rather expectedly, a similar result holds for infima and this proof is left as an exercise. Now we are sure that suprema are unique, how do we prove them? The following theorem provides us with a technique for proving the suprema and infima of specific sets.

Theorem 4.2: Suprema and Infima Analytically

(i) Let $\alpha \in \mathbb{R}$ be an upper bound for a non-empty set $A \subset \mathbb{R}$. Then $\alpha = \sup(A)$ if and only if for every $\varepsilon > 0$ there exists an $a \in A$ such that $\alpha - \varepsilon < a$.

(ii) Let $\beta \in \mathbb{R}$ be a lower bound for a non-empty set $B \subset \mathbb{R}$. Then

$\beta = \inf(B)$ if and only if for every $\varepsilon > 0$ there exists a $b \in B$ such that $b < \beta + \varepsilon$.

Proof. (i) ($\Rightarrow$) Assume $\alpha = \sup(A)$. Then by definition α is the least upper bound for A and hence $\alpha - \varepsilon$ is not an upper bound for A for every $\varepsilon > 0$. Thus, there must exist an $a \in A$ such that $\alpha - \varepsilon < a$.
($\Leftarrow$) Now assume $\alpha \in \mathbb{R}$ is an upper bound for A and that for every $\varepsilon > 0$, there exists an $a \in A$ such that $\alpha - \varepsilon < a$. This means that any number less than α is not an upper bound for A. Thus any other upper bound $b \in \mathbb{R}$ for A must satisfy $\alpha \leq b$. Hence $\alpha = \sup(A)$.

(ii) *Exercise*

□

With this theorem in mind, let's see if we can calculate some infima and suprema explicitly.

Example 4.3: Infima and Suprema

(i) Let $A = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\}$. Then $\sup(A) = 1$.

Proof. Firstly, as $0 = 1 - 1/1$ it follows that $0 \in A$. Hence $A \neq \emptyset$. Now, since $1/n > 0$ for all $n \in \mathbb{N}$, it follows that $1 - 1/n < 1$ for each $n \in \mathbb{N}$. Hence, 1 is an upper bound for A. Now, we need to find a value $N \in \mathbb{N}$, such that

$$1 - \varepsilon < 1 - \frac{1}{N}$$

for all $\varepsilon > 0$. Since the choice of epsilon is arbitrary, we expect the value of N to be a function of epsilon. Rearranging the inequality

$$1 - \varepsilon < 1 - \frac{1}{N}$$

gives

$$N > \frac{1}{\varepsilon}.$$

So we simply need to choose a value of N that is larger than $1/\varepsilon$. How do we know such a value exists? By the Archimedean property Theorem 3.4, we know that such a natural number exists. So, choosing a natural number $N > \frac{1}{\varepsilon}$ gives us the desired property that

$$1 - \varepsilon < 1 - \frac{1}{N}$$

for every $\varepsilon > 0$. □

(ii) Let $B = \{n - 1 : n \in \mathbb{N}\}$. Then $\inf(B) = 0$.

Proof. Clearly B is non-empty. Now, since $n \geq 1$ for each $n \in \mathbb{N}$ it follows that $0 \leq n - 1$ for all $n \in \mathbb{N}$. Hence 0 is a lower bound for B. We now need to find an $N \in \mathbb{N}$ such that

$$N - 1 < 0 + \varepsilon$$

for all $\varepsilon > 0$. Clearly, $N = 1$ satisfies this inequality for all $\varepsilon > 0$. Hence 0 is the greatest lower bound for B. □

There are a couple of things to note from these examples. Firstly, some proofs are clearly tougher than others, even if they are proving similar forms of statements. Unfortunately, there is no algorithmic way to proving all statements, even if they are about the same concept. Additionally, you must keep in mind the theorems that you have proven so far. These can be utilised to prove further theorems and results. They can also greatly simplify your proofs. For example, Example 4.3 (i) utilised the Archimedean property of the real numbers. Without this, the proof would be even more complicated..

You may have also noticed in the examples above that it happened that the supremum or infimum of the set was also contained within the set. If this is the case, these values are called the *maximum* and *minimum* of the set respectively. Note though that the maximum and minimum of set need not always exist as in Example 4.3 (i) and (iii). This emphasises that the maximum and minimum are different to the supremum and infimum, since these must always exist by the completeness axiom. The definition of max and min are explicitly given as follows.

Definition 4.4: Max and Min of Set

Let $A \subset \mathbb{R}$. Then if there exists an $M \in A$ such that $a \leq M$ for all $a \in A$, the set A is said to have a ***maximum*** denoted by $\max(A)$. Similarly, if there exists an element $m \in A$ such that $m \leq a$ for all $a \in A$, the set A is said to have a ***minimum*** denoted by $\min(A)$.

We will finish this chapter by finally proving the very thing that led to us developing the real numbers in the first place. Namely, that there exists a unique positive number $x \in \mathbb{R}$ with $x^2 = 2$.

Example 4.5: $\sqrt{2}$ is a Real Number

Consider the set

$$A = \{x \in \mathbb{R} : x > 0 \text{ and } x^2 < 2\}.$$

Since $A \subset \mathbb{R}$ and it is bounded above by 2, it follows by the completeness axiom that $\alpha := \sup(A) \in \mathbb{R}$ exists. We now want to show that $\alpha^2 = 2$ and $\alpha > 0$. We will do this by showing that if $\alpha^2 < 2$ this leads to a contradiction and similarly if $\alpha^2 > 2$ this also leads to a contradiction. First though, it is easy to establish that $\alpha > 0$ by noticing that $1 \in A$ and $1 \leq \alpha$ implies $\alpha > 0$. Now assume that $\alpha^2 < 2$. The plan is to find an $\varepsilon > 0$ so that

$(\alpha+\varepsilon)^2 < 2$. This would mean that α is not an upper bound for A which is a contradiction. Expanding the bracket above gives

$$(\alpha+\varepsilon)^2 = \alpha^2 + 2\alpha\varepsilon + \varepsilon^2.$$

Choosing $\varepsilon = \min\left\{\frac{1}{2}, \frac{2-\alpha^2}{2\alpha+1}\right\}$ gives

$$(\alpha+\varepsilon)^2 < \alpha^2 + \varepsilon(2\alpha+1) \leq \alpha^2 + \frac{2\alpha+1}{2\alpha+1}(2-\alpha^2) = 2.$$

Hence,we have our desired contradiction. Similarly, now assume that $\alpha^2 > 2$. We now want to find an $\varepsilon > 0$ so that $(\alpha-\varepsilon)^2 > 2$. This would mean that α is not a least upper bound and hence that's the contradiction. Again, expand the bracket to get

$$(\alpha-\varepsilon)^2 = \alpha^2 - 2\alpha\varepsilon + \varepsilon^2.$$

Choosing $\varepsilon = \frac{\alpha^2-2}{2\alpha}$ gives

$$(\alpha-\varepsilon)^2 = 2 + \varepsilon^2 > 2.$$

Again, this is the required contradiction. We then conclude that $\alpha^2 = 2$. Finally, to prove that there is only one positive number that squares to 2, assume there exist $\alpha \neq \tilde{\alpha}$ with $\alpha^2 = \tilde{\alpha}^2$ and $\alpha, \tilde{\alpha} > 0$. Then $\alpha + \tilde{\alpha} > 0$ and thus

$$\begin{aligned} 0 = \alpha^2 - \tilde{\alpha}^2 &= (\alpha - \tilde{\alpha})(\alpha + \tilde{\alpha}) \\ \Rightarrow \alpha - \tilde{\alpha} &= 0 \\ \Rightarrow \alpha &= \tilde{\alpha}. \end{aligned}$$

He who fights monsters should see to it that he himself does not become a monster. And if you gaze for long into an abyss, the abyss gazes also into you.

And that's all folks. This is the end of chapter one on number systems. One final emphasis though on the point that every result that will be proven in this entire course roots from the axioms of the real numbers. Nothing more and nothing less.